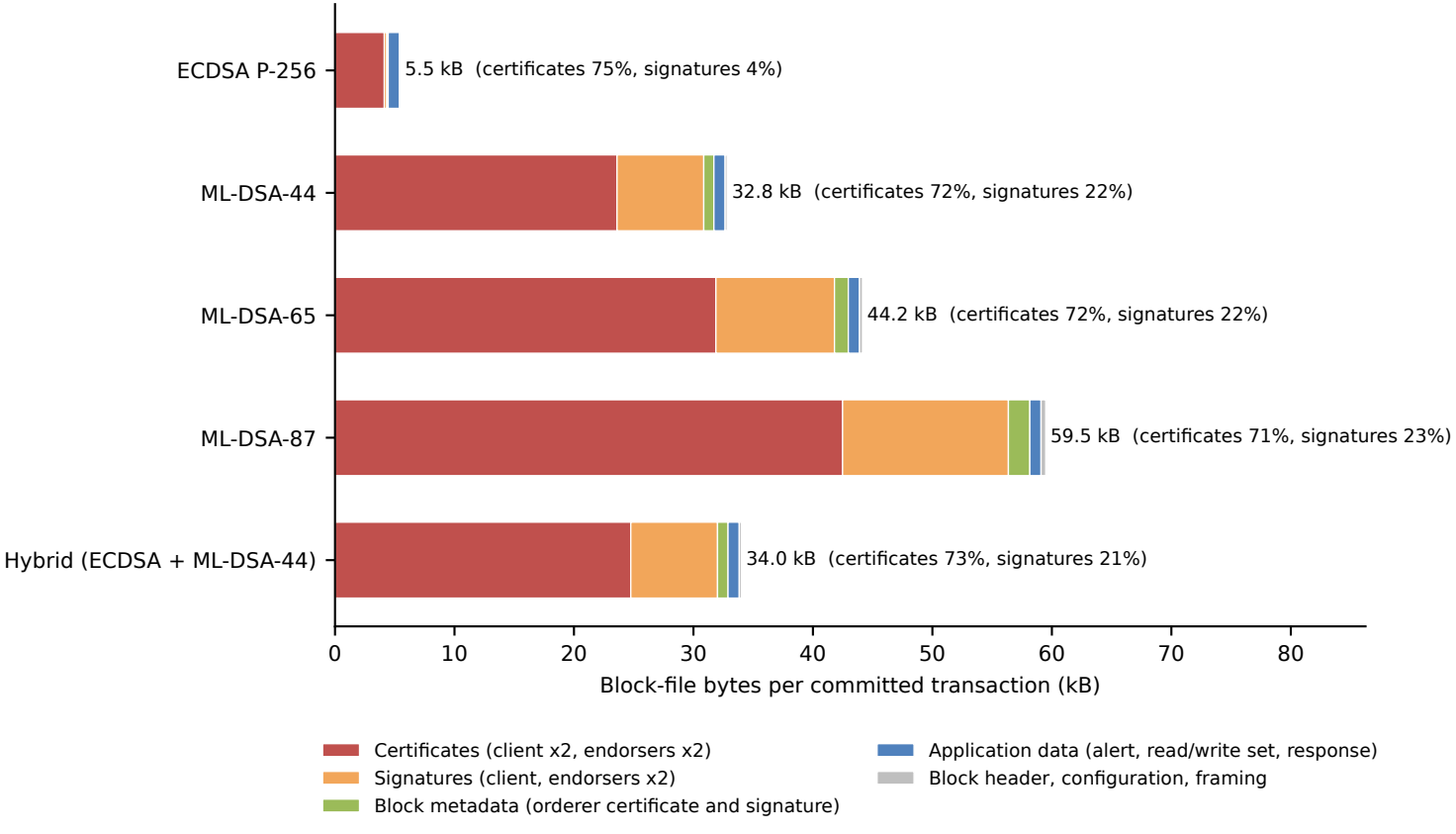
ECDSA P-256
ML-DSA-44
ML-DSA-65
ML-DSA-87
Hybrid (ECDSA + ML-DSA-44)
5.5 kB (certificates 75%, signatures 4%)
32.8 kB (certificates 72%, signatures 22%)
44.2 kB (certificates 72%, signatures 22%)
59.5 kB (certificates 71%, signatures 23%)
34.0 kB (certificates 73%, signatures 21%)
0
10
20
30
40
50
60
70
80
Block-file bytes per committed transaction (kB)
Certificates (client x2, endorsers x2)
Signatures (client, endorsers x2)
Block metadata (orderer certificate and signature)
Application data (alert, read/write set, response)
Block header, configuration, framing